

NETWORK DEVICE DISCOVERY SCANNER – PROJECT SUMMARY

Overview

This project is a Python-based Network Device Discovery Scanner developed to identify active devices connected to a local network.

The tool sends ICMP ping requests to hosts within a specified subnet, detects active devices, and performs reverse DNS lookups to retrieve hostnames. This helps security professionals and network administrators gain visibility into network assets and connected systems.

Features Added

- Automated network host discovery
- ICMP-based device detection
- Reverse DNS hostname resolution
- Professional ASCII banner interface
- Cross-platform compatibility support
- Active device reporting
- Network asset identification
- Exception handling for network errors
- User-friendly terminal output
- Scan summary generation

Information Collected

- IP Address
- Host Availability Status
- Hostname Information
- Total Active Devices
- Network Discovery Results

Technologies Used

- Python
- Subprocess Module
- Socket Programming
- ICMP (Ping)
- DNS Resolution

Example Output

SCAN COMPLETED

Total Active Devices Found: 11

Discovered Hosts:

192.168.1.1	Unknown
192.168.1.102	Unknown
192.168.1.105	Unknown
192.168.1.110	Unknown
192.168.1.111	Unknown
192.168.1.115	Unknown
192.168.1.135	Unknown
192.168.1.139	Unknown
192.168.1.141	Unknown
192.168.1.149	Unknown
192.168.1.152	Unknown

Hostname : Unknown

[+] Device Found

IP Address : 192.168.1.102

Hostname : Unknown

[+] Device Found

IP Address : 192.168.1.105

Hostname : Unknown

[+] Device Found

IP Address : 192.168.1.110

Hostname : Unknown

Concepts Demonstrated

- Network Reconnaissance
- Host Discovery
- Network Enumeration
- ICMP Protocol
- DNS Resolution
- Python Automation
- Network Monitoring
- Asset Discovery

Learning Outcome

This project helps understand how cybersecurity professionals and network administrators discover and monitor devices connected to a network. It provides practical experience with:

- Network Scanning Techniques
- Host Enumeration
- Device Identification
- Network Visibility
- Security Assessment Fundamentals
- Network Asset Management
- Cybersecurity Automation

Security Applications

- Identifying unauthorized devices
- Network inventory management
- Asset discovery and tracking
- Security monitoring
- Internal network assessment
- Infrastructure visibility

Future Enhancements

- Multi-threaded scanning for faster performance
- MAC Address Detection
- Device Vendor Identification
- Open Port Scanning
- Operating System Detection
- CSV/PDF Report Export
- Network Topology Mapping
- Real-Time Monitoring Dashboard

Disclaimer

This project is intended for educational purposes and authorized network assessments only. Always obtain proper permission before scanning or analyzing any network infrastructure.

30 Days of Cybersecurity Automation – Day 07

Building practical cybersecurity tools through hands-on automation and real-world security assessments.

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